Assignment No. 7

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Company maintains employee information such as employee ID, name, designation and salary. Allow users to add, delete information about employees. Display information of a particular employee. If an employee does not exist an appropriate message is displayed. If it is, then the system displays the employee details. Use index sequential files to maintain the data

Code:

```
#include<iostream>
#include<fstream>
#include<iomanip>
#include<cstring>

using namespace std;

struct Employee {
   int id;
   char name[30];
   char designation[30];
   double salary;
};

struct IndexRecord {
   int id;
   int position;
};

class EmployeeSystem {
   string dataFile = "employee.dat";
   string indexFile = "employee.idx";

   public:
   void addEmployee() {
      Employee emp;
      cout << "Enter Employee ID : ";
      cin >> emp.id;

      if (searchIndex(emp.id) != -1) {
         cout << "ERROR: Employee ID already exists\n";
```

```
        return;
    }

    cout << "Enter Name :";
    cin.ignore();
    cin.getline(emp.name, 30);

    cout << "Enter Designation : ";
    cin.getline(emp.designation, 30);

    cout << "Enter Salary : ";
    cin >> emp.salary;

    ofstream outData(dataFile, ios::binary | ios::app);
    int pos = outData.tellp();
    outData.write((char*)&emp, sizeof(Employee));
    outData.close();

    ofstream outIndex(indexFile, ios::binary | ios::app);
    IndexRecord idx = {emp.id, pos};
    outIndex.write((char*)&idx, sizeof(IndexRecord));
    outIndex.close();

    cout << "Employee added successfully !";
}

int searchIndex(int id) {
    ifstream inIndex(indexFile, ios::binary);
    if(!inIndex) return -1;

    IndexRecord idx;
    while(inIndex.read((char*)&idx, sizeof(IndexRecord))) {
        if(idx.id == id && idx.id != -1) {
            int pos = idx.position;
            inIndex.close();
            return pos;
        }
    }
    inIndex.close();
    return -1;
}

void displayEmployee() {
    int id;
    cout << "Enter Employee Id to search : ";
    cin >> id;
    int pos = searchIndex(id);
```

```
        if(pos == -1) {
            cout << "ERROR : Employee wit ID " << id << " does not exists";
        } else {
            ifstream inData(dataFile, ios::binary);
            inData.seekg(pos);
            Employee emp;

            inData.read((char*)&emp, sizeof(Employee));
            cout << "\nEmployee Details:\n";
            cout << "ID: " << emp.id << "\nName: " << emp.name << "\nDesignation: " << emp.designation <<
"\nSalary: " << emp.salary << endl;

            inData.close();
        }
    }

    void deleteEmployee() {
        int id;
        cout << "Enter Employee ID to delete : ";
        cin >> id;

        fstream fileInex(indexFile,ios::binary | ios::in | ios::out);
        IndexRecord idx;
        bool found = false;

        while(fileInex.read((char*)&idx, sizeof(IndexRecord))) {
            if (idx.id == id) {
                idx.id = -1;
                fileInex.seekp((int)fileInex.tellg() - sizeof(IndexRecord));
                fileInex.write((char*)&idx, sizeof(IndexRecord));
                found = true;
                break;
            }
        }
        fileInex.close();

        if(found) cout << "Employee record deleted successfully";
        else cout << "ERROR: Employee ID not found";
    }
};

int main() {

    EmployeeSystem es;
    int choice;

    do {
        cout << "\n------------Employee Management----------\n";
```

```
        cout << "1. Add Employee\n2.Display Employee\n3.Delete Employee\n4.Exit\n";
        cout << "Enter choice : ";
        cin >> choice;

        switch(choice) {
            case 1: es.addEmployee(); break;
            case 2: es.displayEmployee(); break;
            case 3: es.deleteEmployee(); break;
            case 4: cout << "Exiting..."; break;
            default: cout << "Invalid Input"; break;
        }
    } while( choice != 4);

    return 0;
}
```

Output:

```
------------Employee Management----------
1. Add Employee
2.Display Employee
3.Delete Employee
4.Exit
Enter choice : 1
Enter Employee ID : 101
Enter Name :Aditya
Enter Designation : AI Developer
Enter Salary : 150000
Employee added successfully !
------------Employee Management----------
1. Add Employee
2.Display Employee
3.Delete Employee
4.Exit
Enter choice : 1
Enter Employee ID : 102
Enter Name :Parth
Enter Designation : HR
Enter Salary : 90000
Employee added successfully !
------------Employee Management----------
1. Add Employee
2.Display Employee
3.Delete Employee
4.Exit
Enter choice : 1
Enter Employee ID : 103
Enter Name :Rajan
```

```
Enter Designation : Software Developer
Enter Salary : 120000
Employee added successfully !
------------Employee Management----------
1. Add Employee
2.Display Employee
3.Delete Employee
4.Exit
Enter choice : 2
Enter Employee Id to search : 101

Employee Details:
ID: 101
Name: Aditya
Designation: AI Developer
Salary: 150000
------------Employee Management----------
1. Add Employee
2.Display Employee
3.Delete Employee
4.Exit
Enter choice : 1
Enter Employee ID : 104
Enter Name :Kshitij
Enter Designation : Intern
Enter Salary : 15000
Employee added successfully !
```